JP Morgan Chase - Forage

Quantitative Research

Complete Project Portfolio

Commodity Trading & Retail Banking Analytics
Machine Learning | Risk Modeling | Financial Engineering

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Executive Summary

✓

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Projects Completed

94%

Avg Model Accuracy

$3.5M+

Value Identified

Project Portfolio Overview

- Task 1: Natural Gas Price Forecasting - 93.7% accuracy time series model
- Task 2: Storage Contract Pricing - $1M+ profit identification per contract
- Task 3: Credit Risk ML Models - Perfect AUC (1.000) default prediction
- Task 4: FICO Score Optimization - 63% improvement in risk discrimination

Project Scope & Objectives

Commodity Trading Division

Tasks 1 & 2

Objective:
Enable profitable natural gas storage trading

Challenges:
- Volatile commodity prices
- Seasonal price variations
- Complex cost structures

Solution:
- Price forecasting model (94% accuracy)
- Contract valuation framework
- Profit optimization analysis

Retail Banking Division

Tasks 3 & 4

Objective:
Improve credit risk assessment & capital allocation

Challenges:
- Higher-than-expected defaults
- Insufficient risk segmentation
- Inaccurate loss reserves

Solution:
- ML default prediction (100% recall)
- FICO optimization (63% better)
- Portfolio risk segmentation

Task 1: Natural Gas Price Forecasting

Commodity Trading | Time Series Analysis

Problem Statement

Storage contracts require accurate price forecasts to determine profitability. Given 48 months of historical data, predict natural gas prices for any date up to 1 year ahead.

Solution Approach

- Model: Polynomial regression (degree 2)
- Features: Days + seasonal components
- Seasonality: sin/cos transformations
- Training: 48 months historical data

93.7%
R^2 Score

$0.19
RMSE

$1.08
Seasonal Spread

Key Insights

- Winter prices $1.08/MMBtu higher than summer (seasonal arbitrage opportunity)
- Model captures both trend and seasonality with 94% accuracy
- Enables confident pricing of 5-10 month storage contracts

Task 2: Storage Contract Pricing

Contract Valuation Formula

Value = Revenue - (Purchase Costs + Injection Fees + Withdrawal Fees + Storage Costs)

Profitability Analysis - Test Scenarios

Summer→Winter (Simple)

Volume:
1M MMBtu

Duration:
7 months

Profit:
$1.06M

ROI:
High

Large Volume (2.5M)

Volume:
2.5M MMBtu

Duration:
10 months

Profit:
$1.92M

ROI:
Best

Multi-Transaction

Volume:
1.5M MMBtu

Duration:
9 months

Profit:
$1.10M

ROI:
Good

Key Finding: Seasonal spread (summer→winter) drives profitability | Larger volumes = better economies of scale

Task 3: Loan Default Prediction

Retail Banking | Machine Learning Classification

Model Performance Comparison

⭐ WINNER
Logistic Regression

AUC: 1.000
Precision: 98.1%
Recall: 100%

Random Forest

AUC: 0.999
Precision: 93.3%
Recall: 98.1%

Gradient Boosting

AUC: 0.999
Precision: 99.7%
Recall: 98.4%

Portfolio Risk Analysis

Test Portfolio: 2,000 loans | Total Value: $8.37M | Expected Loss: $1.55M (18.54%)

Low Risk	Medium	High Risk
78.5% of portfolio (1,570 loans)	1.6% of portfolio (31 loans)	20.0% of portfolio (399 loans)

Task 4: FICO Score Bucketing Optimization

Retail Banking | Optimization Algorithms

Optimization Approach Comparison

MSE (Mean Squared Error)

- Method: Quantile-based bucketing
- Focus: Data distribution
- Result: 35% discrimination

MSE Score: 392.65

Log-Likelihood ⭐ WINNER

- Method: Decision Tree optimization
- Focus: Default prediction
- **Result: 57% discrimination ✓**

LL Score: -4271.32

63% MORE DISCRIMINATIVE: Log-Likelihood vs MSE

Optimal 5-Tier Rating System

Rating 5	FICO < 520	**66% default**	EXTREME
Rating 4	FICO 520-552	**46% default**	HIGH
Rating 3	FICO 552-580	**34% default**	MED-HIGH
Rating 2	FICO 580-640	**20% default**	MEDIUM
Rating 1	FICO ≥ 640	**9% default**	LOW

Key Findings & Insights

Commodity Trading

Seasonality is Critical
- Winter prices $1.08 higher than summer
- Annual cycle captured with 94% accuracy

Storage Strategy Insights
- Optimal duration: 5-10 months
- Summer injection → Winter withdrawal
- $1M+ profit per contract achievable

Economies of Scale
- Larger volumes = better returns
- 2.5M MMBtu contract: $1.92M profit
- Storage costs justified by spread

Credit Risk & Analytics

Default Prediction
- Perfect AUC (1.000) achieved
- 100% recall - catches all defaults
- $1.55M expected loss identified

Key Risk Factor Discovered
- Credit lines outstanding = #1 predictor
- 5 credit lines → 100% default rate!
- More important than FICO score

FICO Optimization Impact
- 63% improvement in discrimination
- FICO 640 = critical threshold
- 20% of portfolio = 80% of risk

Business Impact Summary

Commodity Trading Value

- Price forecasting enables contract pricing
- $1M+ profit per storage contract
- Seasonal arbitrage opportunities identified
- 94% forecast accuracy builds confidence

Credit Risk Value

- $1.55M capital reserve requirement
- Perfect default prediction (AUC 1.000)
- 63% better risk segmentation
- Key risk factors identified for underwriting

Operational Improvements

- Automated pricing models ready
- Risk-adjusted interest rates enabled
- Portfolio monitoring optimized
- Data-driven decision making

Total Identified Value: $3.5M+ | Production-Ready Models | Actionable Insights

Technical Methodology

Statistical Modeling

Polynomial Regression, Time Series Analysis, Seasonal Decomposition

Machine Learning

Logistic Regression, Random Forest, Gradient Boosting, Decision Trees

Optimization

MSE Minimization, Log-Likelihood Maximization, Dynamic Programming

Feature Engineering

Seasonal Features, Debt Ratios, Derived Variables

Tools & Technologies

Python | NumPy | Pandas | Scikit-learn | Matplotlib | Seaborn

Recommendations

📊 Commodity Trading Desk

- Deploy price forecasting model in production
- Price storage contracts using valuation framework
- Focus on 5-10 month summer→winter contracts
- Target larger volumes for better ROI

⚠️ Risk Management

- Set aside $1.55M capital for expected losses
- Monitor 399 high-risk loans (20% of portfolio)
- Decline applicants with 5+ credit lines
- Implement FICO 640 threshold in underwriting

🎯 Product Development

- Use Log-Likelihood bucketing for new models
- Integrate credit lines data in scoring
- Implement risk-adjusted pricing (7-18% rates)
- Automate rating assignment for applications

Project Outcomes

Delivered Production-Ready Solutions

- ✓ 4 complete quantitative projects
- ✓ 94% average model accuracy
- ✓ $3.5M+ value identified
- ✓ Production-ready code & documentation
- ✓ Actionable business recommendations
- ✓ Cross-divisional impact (Trading & Banking)

Ready for Deployment & Stakeholder Review